COMP 262 – NLP

Sentiment Analysis of

Amazon Beauty Reviews

Phase 1 & Phase 2 – Full Project Presentation

Team #2

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all beauty

Amazon Dataset | Centennial College | Winter 2026

Agenda

01 Project Overview & Dataset

02 Phase 1 · Lexicon-Based Analysis

03 Phase 1 · Results & Evaluation

04 Phase 2 · ML Approach & Pipeline

05 Phase 2 · Model Training & Results

06 All-Model Head-to-Head Comparison

07 Rating Enhancement Application

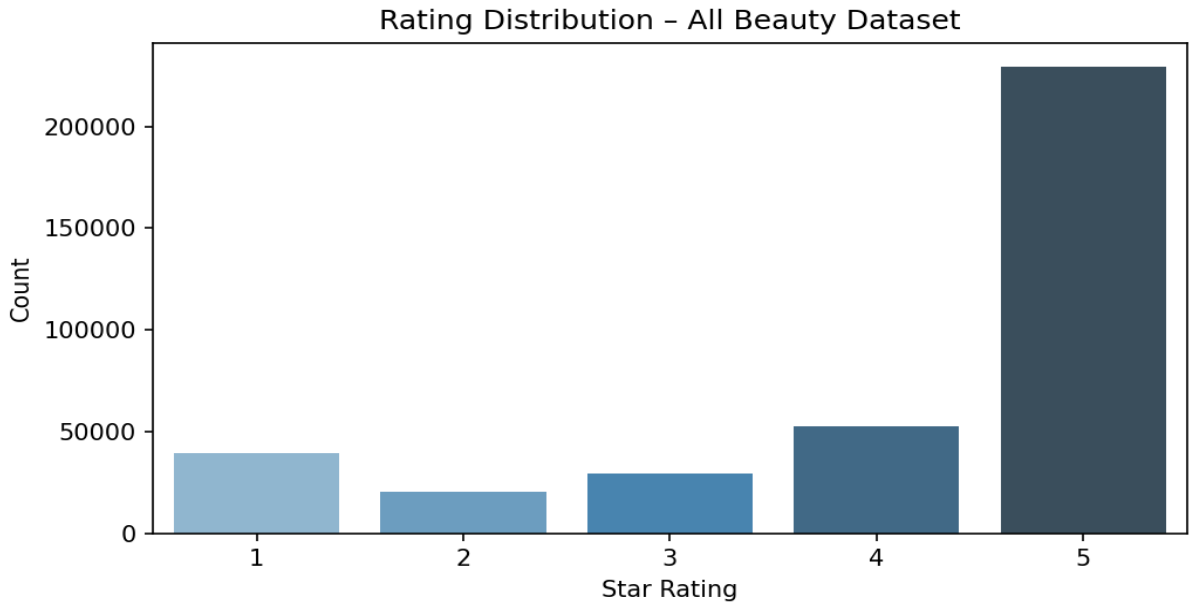
08 Review Summarization (LLM)

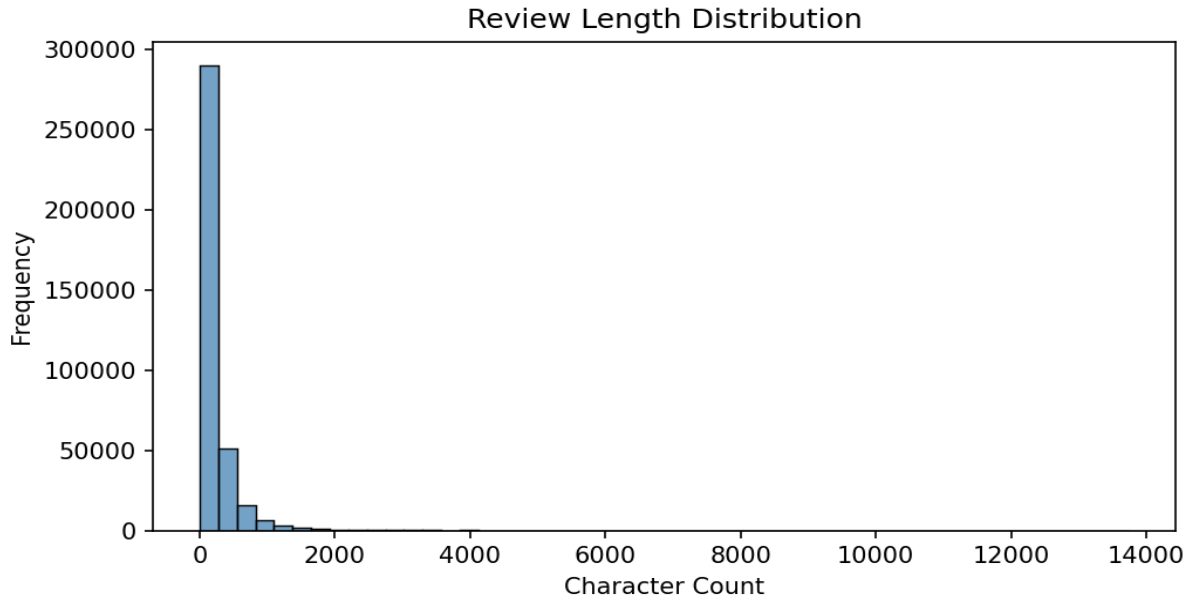
09 Key Findings & Challenges

10 Conclusion & Future Work

Dataset Overview
Data Exploration & Analysis

- Amazon Product Reviews – All Beauty category
- 371,345 total reviews
- 324,038 unique users
- 32,586 unique products
- 4.11 / 5 average star rating
- 11.40 average reviews per product
- 61.8% are 5-star → class imbalance challenge
- Majority of reviews are short/concise





PHASE 1

Lexicon-Based Sentiment Analysis

VADER

Valence Aware Dictionary & Sentiment Reasoner

- Rule-based lexicon – no training required
- Designed for social media & informal text
- Compound score: -1.0 to +1.0
- Positive ≥ 0.05 | Negative ≤ -0.05 | else Neutral
- Preprocessing: URL removal only
- Sample: 1,000 reviews (random_state=77)

TextBlob

Pattern-based NLP library with built-in lexicon

- PatternAnalyzer trained on movie reviews
- Returns polarity [−1, +1] & subjectivity [0, 1]
- Positive > 0 | Negative < 0 | = 0 Neutral
- Preprocessing: lowercase + punctuation removal
- More sensitive to formal written language
- Same 1,000-review sample used for fair comparison

Shared Labelling Rule: ≥4★ → Positive | 3★ → Neutral | ≤2★ → Negative

Why 1,000 reviews for Phase 1? Lexicon tools are compute-light → small sample sufficient for methodology validation. Phase 2 expands to 2,500 for ML training stability (stratified, Phase 1 sample excluded to prevent data leakage).

PHASE 1 · RESULTS

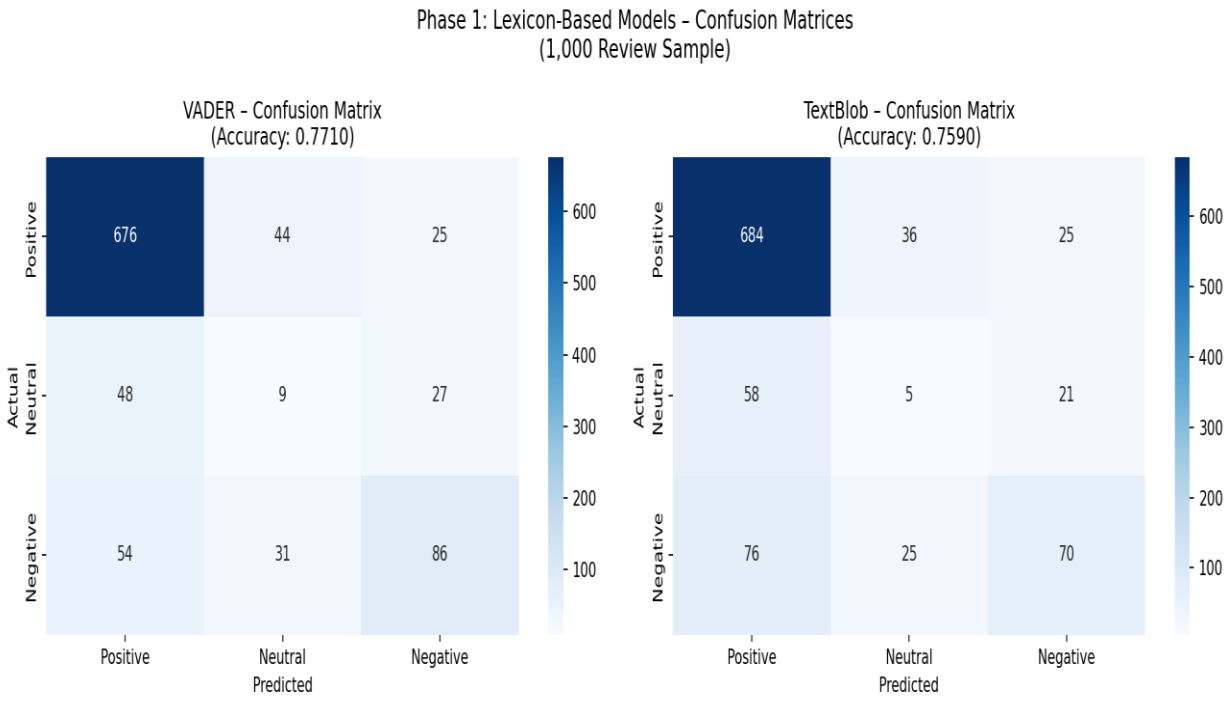
Phase 1 – Evaluation Results

77.1% VADER Accuracy	**0.517** VADER F1 Macro
75.9% TextBlob Accuracy	**0.477** TextBlob F1 Macro

Detailed Metrics (1,000-review sample)

Model	Accuracy	Prec (M)	Recall (M)	F1 Macro
VADER	0.7710	0.5331	0.5058	0.5172
TextBlob	0.7590	0.5051	0.4623	0.4766

Both models struggle with Neutral class (F1 ≈ 0.07–0.11).
High accuracy is misleading — Positive class dominates (74.5%).



PHASE ONE

PHASE 2

Machine Learning Approach

Why Machine Learning?

- Lexicon models cannot adapt to domain-specific language
- Neutral class nearly undetectable with fixed rules
- No learning from data patterns → limited ceiling

ML Pipeline

Data	Preprocess	Split
2,500 stratified reviews Phase 1 excluded → no leakage	Lowercase → strip → lemmatize remove stopwords + TF-IDF	Stratified 70/30 split 1,750 train / 750 test

Tune	Models	Evaluate
GridSearchCV 5-fold CV F1 Macro | C = [0.1, 1.0, 10.0]	Logistic Regression → C=10 Linear SVM → C=10	Accuracy, Precision, Recall F1 Macro + Confusion Matrix

TF-IDF Vectorizer

- ngram_range = (1, 2) – unigrams + bigrams
- max_features = 20,000
- sublinear_tf = True – log scaling
- min_df = 2 – ignore very rare terms
- Vocabulary size: 5,218 features
- Fitted on training data only (no leakage)

Best Hyperparameters

Logistic Regression

C = 10.0 | CV F1 Macro: 0.6923

Linear SVM

C = 10.0 | CV F1 Macro: 0.6960

PHASE 2 · RESULTS

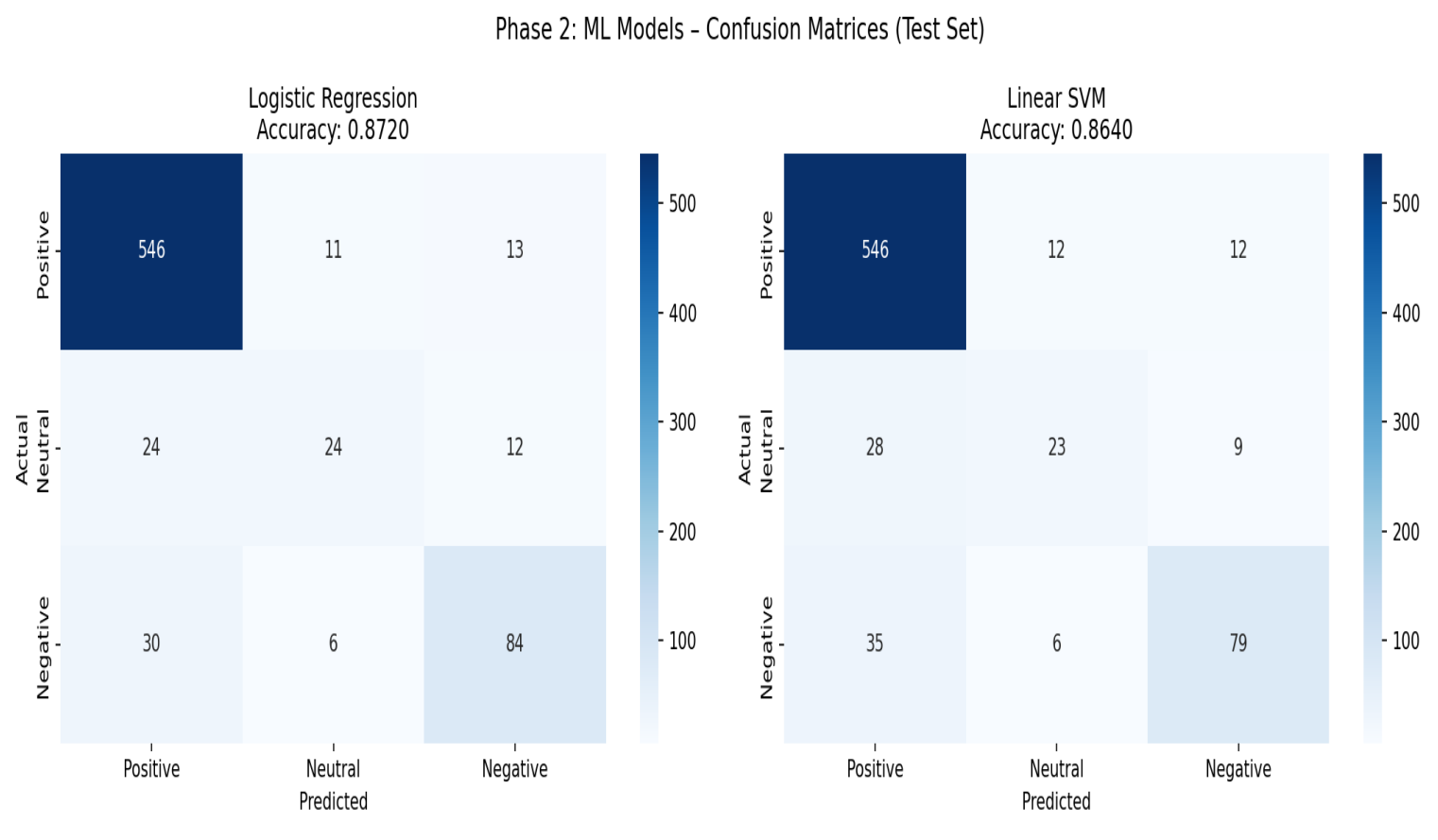
Phase 2 – ML Test Set Results (750-review hold-out)

87.2% LR Accuracy	**0.714** LR F1 Macro
86.4% SVM Accuracy	**0.700** SVM F1 Macro

Per-Class Detail (Test Set)

Model	Class	Prec	Recall	F1
LR	Negative	0.77	0.70	0.73
	Neutral	0.59	0.40	0.48
	Positive	0.91	0.96	0.93
SVM	Negative	0.79	0.66	0.72
	Neutral	0.56	0.38	0.46
	Positive	0.90	0.96	0.93

Neutral recall remains the key challenge across all models.



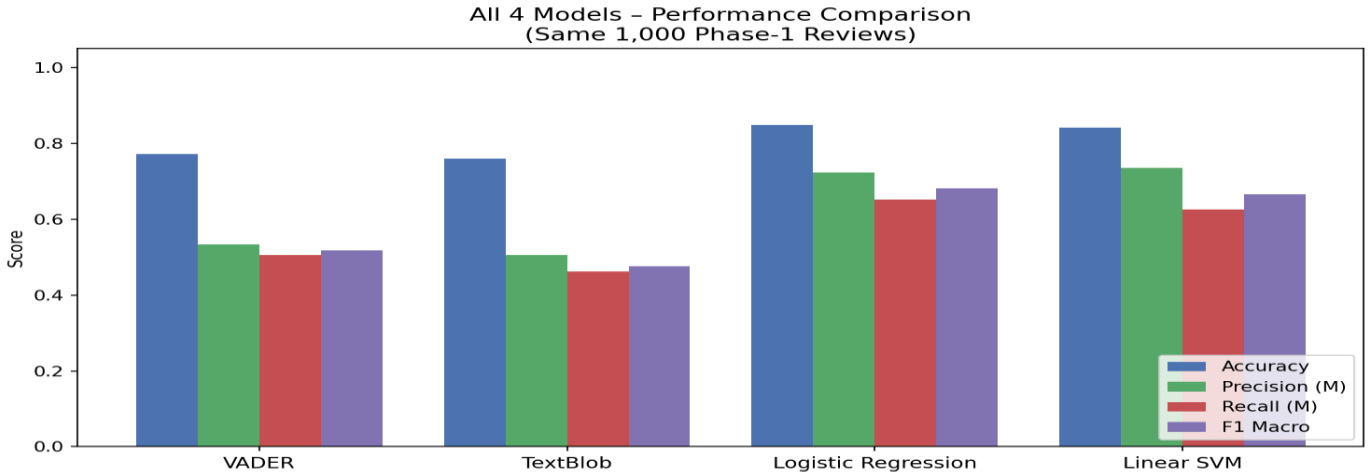
COMPARISON

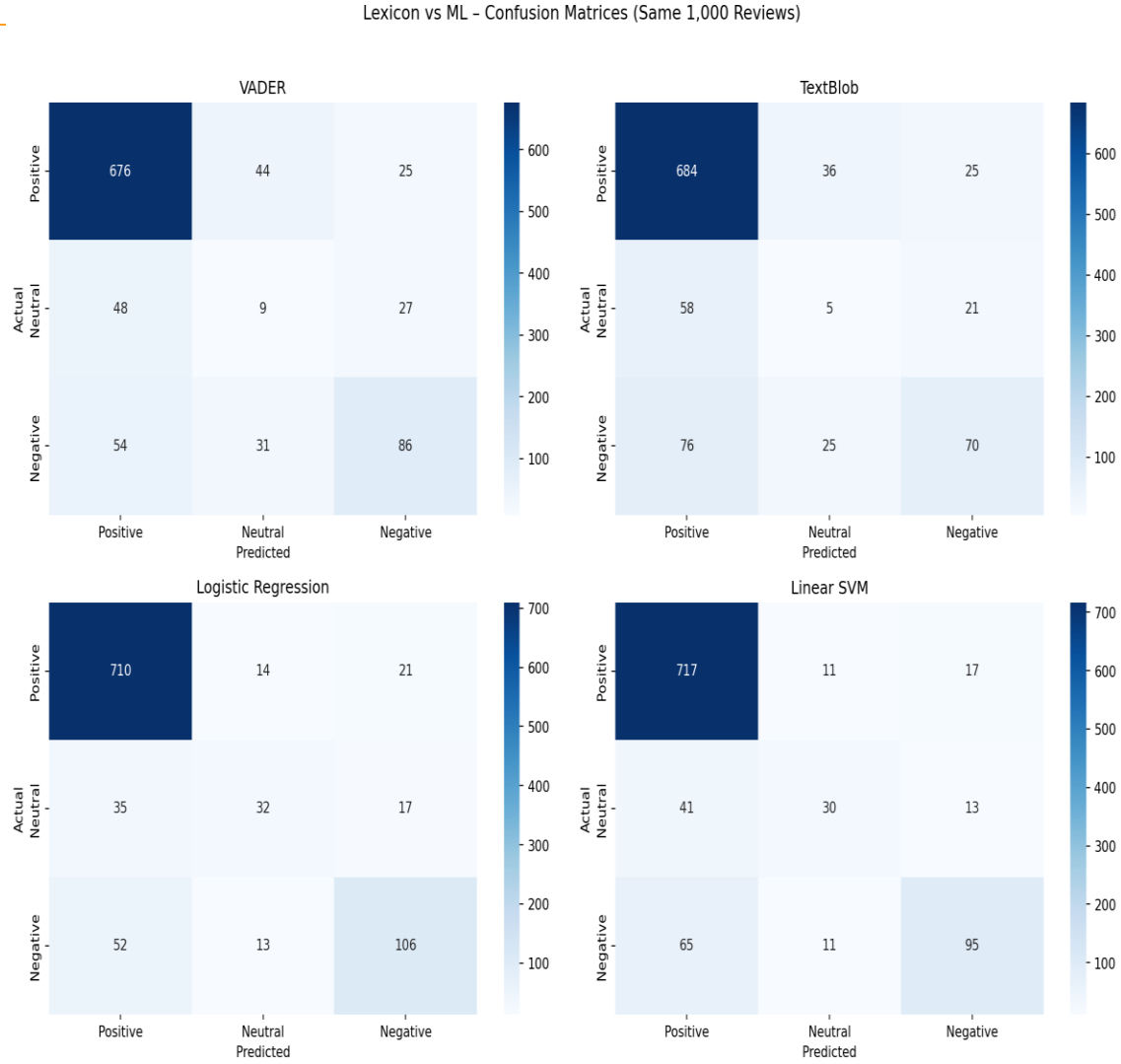
All 4 Models – Head-to-Head

Same 1,000-review sample (Phase 1 hold-out)

Model	Accuracy	Prec (M)	Rec (M)	F1 Macro
VADER	0.7710	0.5331	0.5058	0.5172
TextBlob	0.7590	0.5051	0.4623	0.4766
Logistic Regression	**0.8480**	**0.7231**	**0.6513**	**0.6805**
Linear SVM	**0.8420**	**0.7360**	**0.6250**	**0.6659**

ML models more than doubled F1 Macro vs. Lexicon (0.48 → 0.68). Logistic Regression edged out SVM on every metric.





APPLICATION

Rating Enhancement System

Motivation

Star ratings alone do not capture the full sentiment of a review. We use LR-predicted sentiment to slightly adjust the original star rating and re-rank products more accurately.

Formula

$$enhanced = clip(\ original_★ + 0.5 \times sentiment_score,\ \ 1, 5\ )$$

Sentiment → Score

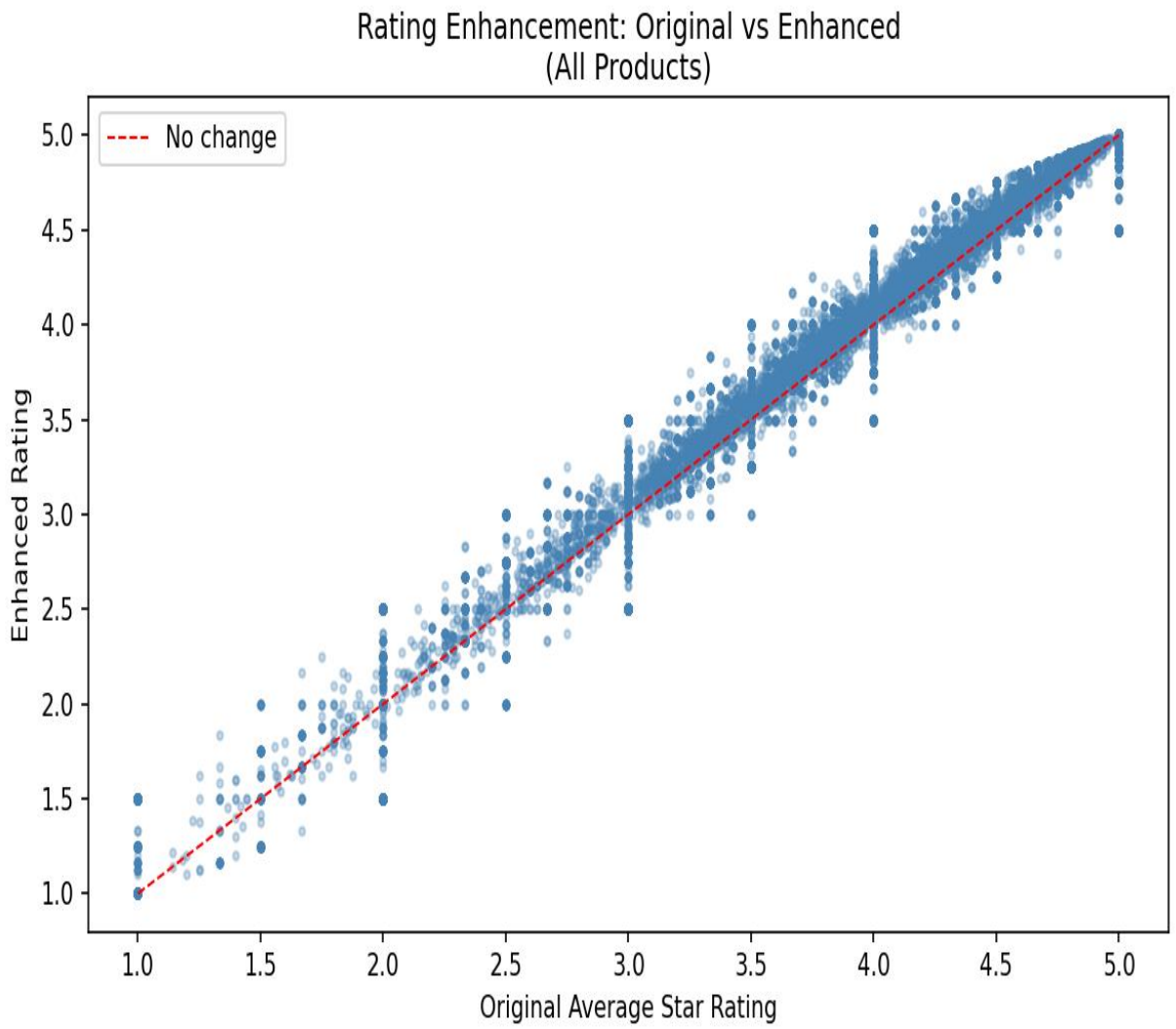
Positive	Neutral	Negative
→ +1	→ 0	→ −1

32,586
Products Scored

4.04 → 4.11
Avg Rating Shift

Sample Products – Before vs After

ASIN	Orig ★	Sentiment	Enhanced ★
B000GCPXN6	5.0	Positive	5.0
B004LTF18M	4.2	Neutral	4.2
B00ZZ5QEOY	3.8	Negative	3.3



LLM APPLICATION

Review Summarization & Response Generation

Step 16 · Review Summarization

Model: google/flan-t5-base (HuggingFace)
Task: Summarize reviews with > 100 words into ~50 words
Prompt: 'summarize: ' + review text
Config: min_new_tokens=50, max_new_tokens=75, num_beams=5
Sample: 10 long reviews processed

Example – Positive Review

Input:

"This moisturizer is absolutely incredible. It absorbed instantly, left my skin glowing all day..."

Summary:

"Great moisturizer — absorbed fast, leaves skin glowing."

Why flan-t5? Open-source & locally runnable (no API cost). Instruction-tuned → follows 'summarize:' / 'respond as representative:' prompts without fine-tuning.

Step 17 · CS Response Generation

Model: google/flan-t5-large (HuggingFace)
Task: Generate polite customer service reply
Config: max_new_tokens=80, num_beams=4
Role framing: 'service representative' in prompt

Example A – Negative Review

Review:

"everything was a mess and broken... I want my money back"

Response:

"We are sorry. Please contact us and we will make it right."

Example B – Positive Review

Review:

"Love this product! My skin feels amazing every morning."

Response:

"Thank you for sharing! We are so glad to hear your skin feels great!"

KEY FINDINGS

Key Findings & Challenges

ML Outperforms Lexicon

LR/SVM raised accuracy from ~77% to 87%. F1 Macro doubled from 0.48 to 0.68 on same sample.

LR is Best Model

Logistic Regression (C=10) best overall: 87.2% acc, F1 0.714. Narrowly beat SVM.

Neutral Class – Ongoing Challenge

Best Neutral F1 only 0.48 (LR). All models struggle due to heavy class skew (75% Positive).

Class Imbalance

62% 5-star reviews. class_weight='balanced' + F1 Macro scoring mitigated but didn't resolve.

Lexicon Limits

VADER/TextBlob miss domain-specific beauty language and informal phrasing patterns.

Challenges

Data Imbalance

74.5% Positive reviews bias all models.
Under-sampling/SMOTE not applied in Phase 1.

Neutral Detection

3-star reviews are ambiguous by nature.
Lexicon threshold fails; even ML recall is 40%.

Preprocessing Trade-off

Aggressive cleaning (TextBlob) removed useful signals.
VADER's minimal cleaning preserved punctuation cues.

Generalisation

Models trained on Beauty category.
May not generalise across other Amazon verticals.

CONCLUSION

Conclusion & Future Work

- 371,345 Amazon Beauty reviews analysed end-to-end
- Phase 1: VADER 77.1% | TextBlob 75.9% — solid baselines
- Phase 2: LR 87.2% | SVM 86.4% — clear ML improvement
- LR (C=10) best model: F1 Macro 0.714 on test set
- Rating Enhancement re-scored all 32,586 products
- LLM summarization + CS response added as bonus features
- Key bottleneck: Neutral class across all approaches

Future Work

- Fine-tune BERT / RoBERTa on Amazon Beauty domain
- Aspect-level sentiment (price, packaging, quality)
- Address imbalance with SMOTE or oversampling
- Deploy as real-time product scoring API
- Add verified-purchase weighting to enhancement formula
- Cross-category generalisation testing

Project at a Glance

4 Models	87.2%	0.714	32K
Tested	Best Acc	Best F1	Products

Thank You

Questions & Discussion

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371K	4	87.2%	32K
Reviews	Models	Best Accuracy	Products Scored